5.3 Operating conditions

5.3.1 General operating conditions

Table 24. General operating conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V_{DD}	Standard operating voltage	-	2.0(1)	3.6	V
V_{IN}	I/O input voltage	-	-0.3	Min (V_{DDIO1} + 3.6, 5.5)(2)	V
f_{HCLK}	AHB clock frequency	-	-	48	MHz
f_{PCLK}	APB clock frequency	-	-	48	MHz
T_A	Ambient temperature(3)	Suffix 6(4)	-40	85	°C
		Suffix 7(4)	-40	105	
		Suffix 3(4)	-40	125	
T_J	Junction temperature	Suffix 6(4)	-40	105	°C
		Suffix 7(4)	-40	125	
		Suffix 3(4)	-40	130	

1. When RESET is released, functionality is guaranteed down to V_{PDR}.
2. For operation with voltage higher than VDD +0.3 V, the internal pull-up and pull-down resistors must be disabled.
3. The T_A(max) applies to P_D(max). At $P_D < P_D$(max) the ambient temperature is allowed to go higher than T_A(max) provided that the junction temperature T_J does not exceed T_J(max). Refer to *Section 6.9: Thermal characteristics*.
4. Temperature range digit in the order code. See *Section 7: Ordering information*.

5.3.2 Operating conditions at power-up / power-down

The parameters given in *Table 25* are derived from tests performed under the ambient temperature condition summarized in *Table 24*.

Table 25. Operating conditions at power-up / power-down

Symbol	Parameter	Min	Max	Unit
t_{VDD}	V_{DD} rise time rate	0	∞	µs/V
	V_{DD} fall time rate	10	∞	

5.3.3 Embedded reset and power control block characteristics

The parameters given in *Table 26* are derived from tests performed under the ambient temperature conditions summarized in *Table 24*.

Table 26. Embedded reset and power control block characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{RSTTEMPO}$(1)	POR temporization when V_{DD} crosses V_{POR}	V_{DD} rising	-	270	500	µs
V_{POR}(1)	Power-on reset threshold	-	1.9	1.94	1.98	V